

Project Report

on

“Research and Design of a Wearable Device for Real-Time Carbohydrate Intake Monitoring using Machine Learning”

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Under the guidance

of

Prof. Pallavi Nehete

In partial fulfilment of

Integrated B.Tech. Computer Science and Engineering

(Third Year)

[2024-2025]

At



Dr. Vishwanath Karad

**MIT WORLD PEACE
UNIVERSITY** | PUNE

TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Certificate of Completion

This is to certify that **Mr. Atharva Magdum, Mr. Piyush Harpude, Mr. Anish Erande, Mr. Yash Wahane** from DVK's, Department of Polytechnic having **PRN No. 1032220796, 1032220794, 1032220797, 1032220777** have successfully completed project of Third year having title **“Research and Design of a Wearable Device for Real-Time Carbohydrate Intake Monitoring using Machine Learning”** under guidance of **Prof. Pallavi Nehete** The project is being submitted towards partial fulfilment of Integrated B.Tech third year Program in Computer Science and Engineering for academic year 2024-25.

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Place: MIT-WPU

ACKNOWLEDGEMENT

We are profoundly grateful to **Prof. Pallavi Nehete** of Computer Engineering Department for her expert guidance and continuous encouragement throughout to see that this project meets its target since its commencement to its completion. We would like to express our deepest appreciation towards **Dr. Prof. R. S. Kale (Program Director), Prof. Jyoti Mante (Program Head)** of Computer Science and Engineering of **Dr. Vishwanath Karad MIT World Peace University, Department of Polytechnic Pune**, whose invaluable guidance supported me in completing this project.

At last, we must express our sincere heartfelt gratitude to all the staff members of the Computer Science and Engineering Department who helped us directly or indirectly during this course of work.

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ABSTRACT

Metabolic conditions such as diabetes, obesity, and cardiovascular diseases continue to rise globally, driven largely by excessive carbohydrate intake and fluctuating glycemic levels. Traditional methods for monitoring glycemic response, such as invasive continuous glucose monitors (CGMs) and manual food logs, are often inaccessible, inconvenient, or prone to inaccuracies. In this research-based project, we propose and develop a non-invasive, machine learning-powered predictive model that estimates daily carbohydrate intake and predicts real-time glycemic response based on biometric data collected from existing wearable devices.

The model leverages advanced ensemble learning techniques including XGBoost, Random Forest, and LSTM to process features such as heart rate variability (HRV), SpO₂ levels, body temperature, and physical activity — data commonly available from commercial wearables. This model can be integrated into existing wearable platforms or mobile health applications to deliver personalized dietary feedback. Experimental validation shows a strong performance, achieving an R² score of 0.87 and RMSE of 3.92. The proposed solution offers a scalable, affordable, and non-invasive approach to metabolic health monitoring without requiring custom hardware.

Keywords: *Carbohydrate monitoring, glycemic prediction, non-invasive sensing, metabolic health;*

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